Strength properties

- $S_{Lt} = longitudinal\ tensile\ strength$
- $S_{Tt} = transverse\ tensile\ strength$
- $S_{Lc} = longitudinal\ compressive\ strength$
- $S_{Tc} = transverse\ compressive\ strength$
- $S_{Lts} = in - plane\ shear\ strength$
- Nomenclature may vary depending on textbook

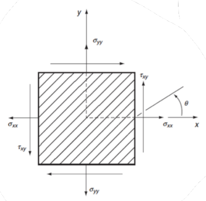
Failure criterion – what do you consider failure?

- Maximum stress theory
- Maximum strain theory
- Tsai-Hill failure theory
- Tsai-Wu theory
- To use them, applied stresses (or strains) are first transformed into principal material directions
- The transformed stresses are denoted $\sigma_1, \sigma_2, \tau_{12}$ and the applied stresses are denoted σ_x, σ_y, τ_{xy}

$$\begin{bmatrix}\sigma_1\\ \sigma_2\\ \tau_{12}\end{bmatrix} = \begin{bmatrix}\cos^2\theta & \sin^2\theta & 2\sin\theta\cos\theta\\ \sin^2\theta & \cos^2\theta & -2\sin\theta\cos\theta\\ -\sin\theta\cos\theta & \sin\theta\cos\theta & \cos^2\theta-\sin^2\theta\end{bmatrix}\begin{bmatrix}\sigma_x\\ \sigma_y\\ \tau_{xy}\end{bmatrix} = [\mathrm{T}]\begin{bmatrix}\sigma_x\\ \sigma_y\\ \tau_{xy}\end{bmatrix}$$

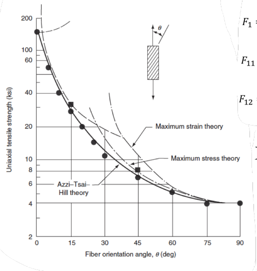
Maximum stress theory

- Failure occurs when any stress in the principal material directions is equal to or greater than the corresponding ultimate strength.
- Thus, to avoid failure, the following criterion need to be met:
  - $\sigma_1 < S_{Lt}$
  - $\sigma_2 < S_{Tt}$
  - $\tau_{12} < S_{Lts}$
  - Assume compressive and tensile strength are the same



Simple case of uniaxial tensile in *x*-direction

- For the simple case of uniaxial tensile loading in the *x*-direction, only σ_x is present and $\sigma_y = \tau_{xy} = 0$
- The transformed stresses are
  - $\sigma_1 = \sigma_x \cos^2\theta$
  - $\sigma_2 = \sigma_x \sin^2\theta$
  - $\tau_{12} = -\sigma_x \sin\theta\cos\theta$



Critical values of fiber orientation

- Using the maximum stress theory, failure of the lamina is predicted if the applied stress σ_x exceeds the smallest (in absolute value) of
  - $\sigma_x = S_{Lt}/\cos^2\theta$
  - $\sigma_x = S_{Tt}/sin^2\theta$
  - $\sigma_x = S_{Lts}/-\sin\theta\cos\theta$
- Thus, the safe value of σ_x also depends on the fiber orientation angle θ and not just the strength values
- At small values of θ, longitudinal tensile failure is expected; at high values, transverse tensile failure is expected; at intermediate values, in-plane shear failure of the lamina is expected

Maximum strain theory

- Failure occurs when any strain in the principal material directions is equal to or greater than the corresponding ultimate strain
- Thus, to avoid failure, following criterion need to be met:
  - $\epsilon_1 < \epsilon_{Lt}$
  - $\epsilon_2 < \epsilon_{Tt}$
  - $\gamma_{12} < \gamma_{LTs}$

Simple case of uniaxial tensile in *x* direction

- For the simple case of uniaxial tensile loading in the x-direction, only σ_x is present and $\sigma_y = \tau_{xy} = 0$
- Using the stress-strain relationship, and the transformed stresses, the strains in the principal material directions are
  - $\epsilon_1 = S_{11}\sigma_1 + S_{12}\sigma_2 = (S_{11}\cos^2\theta + S_{12}\sin^2\theta)\sigma_x$
  - $\epsilon_2 = S_{12}\sigma_1 + S_{22}\sigma_2 = (S_{12}\cos^2\theta + S_{22}\sin^2\theta)\sigma_x$
  - $\gamma_{12} = S_{66}\tau_{12} = -S_{66}\sin\theta\cos\theta\,\sigma_x$

Simple case of uniaxial tensile in *x* direction

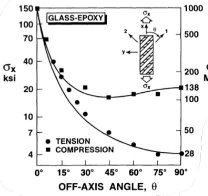
- Using the maximum strain theory, failure of the lamina is predicted if the applied stress σ_x exceeds the smallest of
  - $\dfrac{\epsilon_{Lt}}{S_{11}\cos^2\theta + S_{12}\sin^2\theta} = \dfrac{E_{11}\epsilon_{Lt}}{\cos^2\theta - \nu_{12} sin^2\theta} = \dfrac{S_{Lt}}{\cos^2\theta - \nu_{12} sin^2\theta}$
  - $\dfrac{\epsilon_{Tt}}{S_{12}\cos^2\theta + S_{22}\sin^2\theta} = \dfrac{E_{22}\epsilon_{Tt}}{\sin^2\theta - \nu_{21}\cos^2\theta} = \dfrac{S_{Tt}}{sin^2\theta - \nu_{21}\cos^2\theta}$
  - $\dfrac{\gamma_{LTs}}{S_{66}\ sin\,\theta\ cos\,\theta} = \dfrac{G_{12}\gamma_{LTs}}{sin\,\theta\ cos\,\theta} = \dfrac{S_{LTs}}{sin\,\theta\ cos\,\theta}$
- Main difference between the maximum strain failure criterion and maximum stress failure criterion is the inclusion of Poisson's ratio

Tsai-Hill theory

- Failure occurs in an orthotropic lamina if and when the following equality is satisfied:
  - $\dfrac{\sigma_1^2}{S_{Lt}^2} - \dfrac{\sigma_1\sigma_2}{S_{Lt}^2} + \dfrac{\sigma_2^2}{S_{Tt}^2} + \dfrac{\tau_{12}^2}{S_{Lts}^2} = 1$
- σ_1 and σ_2 are both tensile (positive) stresses
- When σ_1 and σ_2 are compressive, the corresponding compressive strengths are used
- Substitution of stress-transformation equations (for uniaxial tensile stress situation)
  - $\sigma_1 = \sigma_x \cos^2\theta$
  - $\sigma_2 = \sigma_x \sin^2\theta$
  - $\tau_{12} = -\sigma_x \sin\theta\cos\theta$

Tsai-Hill theory

- For the uniaxial tensile stress situation considered earlier, failure is predicted if
  - $\sigma_x \geq \dfrac{1}{\left(\frac{\cos^4\theta}{S_{Lt}^2} - \frac{\sin^2\theta\cos^2\theta}{S_{Lt}^2} + \frac{\sin^4\theta}{S_{Tt}^2} + \frac{\sin^2\theta\cos^2\theta}{S_{Lt}^2}\right)^{\frac{1}{2}}}$
- This equation indicates a better match with experimental data than the maximum stress or the maximum strain theories
- Only has one criterion, not three as in previous failure criteria



Tsai-Wu failure theory

- The Tsai-Wu failure criterion is a material failure theory that is widely used for anisotropic composite materials.
- The basic assumption of the criterion is that there exists a failure surface in the stress-space in the scalar form of

$$\Sigma_{i=1}^{6} F_i\sigma_i + \Sigma_{i=1}^{6}\Sigma_{j=1}^{6} F_{ij}\sigma_i\sigma_j = 1$$

- F_i and F_{ij} are strength coefficients determined from material strength experiments.
- σ_i are the stress values in contracted notation.

Tsai-Wu failure theory

- Under plane stress conditions, the Tsai-Wu failure theory for an orthotropic lamina gives:

$$F_1\sigma_1 + F_2\sigma_2 + F_6\sigma_6 + F_{11}\sigma_1^2 + F_{22}\sigma_2^2 + F_{66}\sigma_6^2 + 2F_{12}\sigma_1\sigma_2 = 1$$

$$F_1 = \frac{1}{\sigma_{1t}} - \frac{1}{\sigma_{1c}} \qquad F_2 = \frac{1}{\sigma_{2t}} - \frac{1}{\sigma_{2c}} \qquad F_6 = 0$$

$$F_{11} = \frac{1}{\sigma_{1c}\sigma_{1t}} \qquad F_{22} = \frac{1}{\sigma_{2c}\sigma_{2t}} \qquad F_{66} = \frac{1}{S^2}$$

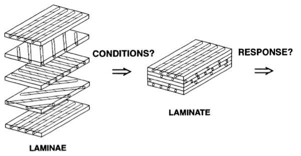
$$F_{12} = \frac{1}{2\sigma_b^2}[1 - (F_1 + F_2)\sigma_b - (F_{11} + F_{22})\sigma_b^2]$$

Symbol	Variable name (magnitude)
σ_{1t}	Longitudinal tensile strength
σ_{1c}	Longitudinal compressive strength
σ_{2t}	Transverse tensile strength
σ_{2c}	Transverse compressive strength
S	Shear failure strength
σ_b	Failure strengths in equibiaxial tension

Difficulties with failure criteria

- For composite materials, many interactive failure criteria have been proposed, and none is universally accepted as the best
- One of the difficulties in coming up with a criterion that works for all materials and stress stages in that failure can take many forms. A layer may fail due to fiber breaking, resin cracking, or fiber buckling.
- Also, for some applications the failure of a single layer may not result in ultimate failure of the laminate
- For example, in a composite pressure vessel the resin between fibers will typically crack at a pressure much less than than the ultimate failure pressure.

Behavior of a laminate



- A laminate is constructed by stacking a number of laminas in the thickness (*z*) direction
- How will a laminate respond to loading (imposed forces and moments)?
- Textbook Jones Chapter 4

From lamina to laminate

- Examples of a few special types of laminates
  - Unidirectional: Fiber orientation angles are the same in all laminas. In unidirectional 0° laminates, $\theta = 0°$ in all laminas
  - Angle-ply: Fiber orientation angles in alternate layers are $\theta/-\theta/\theta/-\theta$ when θ does not equal 0° or 90°
  - Cross-ply: Fiber orientation angles in alternate layers are $0/90/0/90$
  - Symmetric laminate: The ply orientation is symmetrical about the centerline of the laminate; that is, for each ply above the midplane, there is an identical ply (in material, thickness, and fiber orientation angle) at an equal distance below the midplane. For a symmetric laminate, $\theta(\mathrm{z}) = \theta(-\mathrm{z})$, where z is the distance from the midplane of the laminate

Symmetric laminate examples and codes

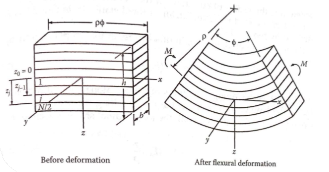
- $[0/45/-45/45/-45/45/-45/0/0/0/0/0/-45/45/-45/45/-45/45/0]$ → $[0/(\pm45)_3/0_2/\bar{0}]_S$
- $[\theta/-\theta/\theta/-\theta/-\theta/\theta/-\theta/\theta]$ → $[\theta/-\theta]_{2S}$ or $[\pm\theta]_{2S}$
  - Two adjacent $\pm\theta$ plies on each side of the plane of symmetry are denoted by the subscript 2S
- Symmetric angle-ply laminate: $[\theta/-\theta/\theta/-\theta/\theta/-\theta/\theta]$ → $[\pm\theta/\theta/-\bar{\theta}]_S$
  - Symmetric angle-ply laminates contain an odd number of plies
- Symmetric cross−ply laminate: $[0/90/0/90/0/90/0/90/0/]$ → $[(0/90)_2/\bar{0}]_S$
- Hybrid laminate: $[0_B/0_B/45_C/-45_C/90_G/90_G/-45_C/45_C/0_B/0_B]$ → $[0_{2B}/(\pm45)_C/90_G]_S$
  - where B, C, and G represent boron, carbon, and glass fiber

Other types of laminates

- Anti-symmetric laminate: the ply orientation is antisymmetric about the centerline of the laminate; that is, for each ply of fiber orientation angle θ above the midplane, there is a ply of fiber orientation angle $-\theta$ with identical material and thickness at an equal distance below the midplane
  - Example: $[\theta/-\theta/\theta/-\theta]$
- Unsymmetric laminate: No symmetry or antisymmetry
  - Examples: $[0/0/0/90/90/90]$ and $[0/45/-45/90]$

Theory of laminated beams in pure flexure

- Simplest laminated structure is a laminated beam that is subjected to pure bending
- A theory of laminated beams in pure flexure can be developed from mechanics of materials concepts



A section of a rectangular laminated beam of depth h and width b

Assumptions

- Plane sections that are initially normal to the longitudinal axis of the beam remain plane and normal during flexure
- The plies are perfected bonded together so that no slip occurs at ply interfaces
- The beam has both geometric and material property symmetry about the neutral surface (the plies are symmetrically arranged about the *xy* plane)
- Each ply is linearly elastic with no shear coupling (ply orientations are either 0 or 90 degrees)

Theory of laminated beams in pure flexure

- Longitudinal normal strain at a distance *z* from the neutral surface
  - $\epsilon_x = \dfrac{(\rho+z)\phi - \rho\phi}{\rho\phi} = \dfrac{\mathrm{z}}{\rho}$
  - ρ is the radius of curvature of the neutral surface during flexure, ϕ the angle, and *z* the distance from neutral surface defined by the *xy* plane
- Using assumption 4, the longitudinal stress in the *j*th ply is given by
  - $(\sigma_x)_j = (E_x)_j(\epsilon_x)$
  - $(E_x)_j$ is the Young's modulus of jth ply along the *x* direction and (ϵ_x) is the longitudinal strain in the jth ply along the x direction
  - → $(\sigma_x)_j = (E_x)_j(\epsilon_x) = (E_x)_j\,\dfrac{\mathrm{z}}{\rho}$

Theory of laminated beams in pure flexure

- Static equilibrium requires (and symmetry assumption)
  - $M = \int dM = \int zdF = \int z\sigma_x dA = \int z\sigma_x bdz = 2\int_0^{\frac{h}{2}} \sigma_x zbdz$
  - $M = 2\int_0^{\frac{h}{2}} \sigma_x zbdz = 2\sum_{j=1}^{\frac{N}{2}} \int_{z_{j-1}}^{z_j} \sigma_x zbdz = 2\sum_{j=1}^{\frac{N}{2}} (E_x)_j \frac{\mathrm{b}}{\rho} \int_{z_{j-1}}^{z_j} z^2 dz$
  - $M = \frac{2b}{\rho}\sum_{j=1}^{\frac{N}{2}} (E_x)_j \frac{\left(z_j^3 - z_{j-1}^3\right)}{3} = \frac{2b}{3\rho}\sum_{j=1}^{\frac{N}{2}} (E_x)_j\,(z_j^3 - z_{j-1}^3)$
  - *N* is the total number of plies and z_j is the distance from the neutral surface to the outside of the *j*th ply
- For an even number of plies of uniform thickness $z_j = \frac{jh}{N}$
  - $M = \frac{2bh^3}{3\rho N^3}\sum_{j=1}^{\frac{N}{2}} (E_x)_j\,(3j^2 - 3j + 1)$

Theory of laminated beams in pure flexure

- For a homogeneous, isotropic beam, the moment-curvature relation
  - $M = \frac{E_{fl}I_{yy}}{\rho} = \frac{E_{fl}bh^3}{12\rho}$, $I_{yy} = \frac{bh^3}{12}$
  - E_{fl} is the effective flexural modulus of the beam
- Combining equations we find the effective flexural modulus of the laminated beam
  - $E_{fl} = \frac{8}{h^3}\sum_{j=1}^{\frac{N}{2}} (E_x)_j\,(z_j^3 - z_{j-1}^3)$ (even number of plies)
  - $E_{fl} = \frac{8}{N^3}\sum_{j=1}^{\frac{N}{2}} (E_x)_j\,(3j^2 - 3j + 1)$ (even number of plies with uniform thickness)
- The flexural modulus of the laminated beam depends on the ply-stacking sequence and the ply moduli

Theory of laminated beams in pure flexure

- The deflections of laminated beams can now be calculated by using the flexural modulus
- Differential equation for the transverse deflection, *w*, of a beam
  - $E_{fl}I_{yy}(d^2w/dx^2) = M$
  - $w_{max} = \frac{PL^3}{3E_{fl}I_{yy}}$ at x = L (maximum deflection at the tip of a laminated cantilever beam)
  - P is the applied tip load and L is the beam length
- The Euler buckling load, P_{cr}, for a laminated beam
  - $P_{cr} = \frac{\pi^2 E_{fl}I_{yy}}{L_e^2}$
  - L_e is the effective length that includes end effect

Theory of laminated beams in pure flexure

- An alternative expression for the stress in the *j*th ply can be
  - $(\sigma_x)_j = \frac{M}{E_{fl}I_{yy}}(E_x)_j\,z = \frac{Mz}{I_{yy}}\left[\frac{(E_x)_j}{E_{fl}}\right]$
  - The term in square brackets can be thought of as a correction term, which when multiplied by the familiar homogeneous isotropic beam stress, gives the stress in the *j*th ply of the laminated beam
  - Another important thing to note is that the maximum stress in the laminated beam does not always occur on the outer surface as it does in the homogeneous, isotropic beam
  - At each section in a laminated beam, the ratio $\frac{M}{E_{fl}I_{yy}}$ is constant, and the remaining term $(E_x)_j\,z$ determines the maximum stress.

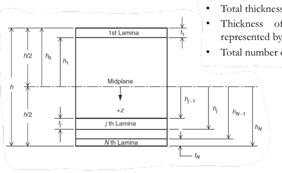
Lamination theory

- Lamination theory is useful in calculating stresses and strains in each lamina of a thin laminated structure
- Beginning with the stiffness matrix of each lamina, the step by step procedure in lamination theory includes
  - Calculation of stiffness matrices for the laminate
  - Calculation of midplane strains and curvatures for the laminate due to a given set of applied forces and moments
  - Calculation of in-plane strains $\epsilon_{xx}, \epsilon_{yy}, \gamma_{xy}$ for each lamina
  - Calculation of in-plane stresses $\sigma_{xx}, \sigma_{yy}, \tau_{xy}$ for each lamina

Basic assumptions in the lamination theory

- Laminate is thin and wide (width >> thickness)
- A perfect interlaminar bond exists between various laminas
- Strain distribution in the thickness direction is linear
- All laminas are macroscopically homogeneous and behave in a linearly elastic manner

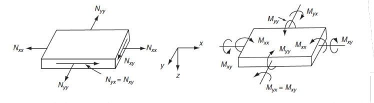
- Geometric midplane of the laminate contains the *xy* axes and the *z* axis defines the thickness direction
- Total thickness of the laminate is *h*
- Thickness of various laminas are represented by t_1, t_2, t_3, and so on
- Total number of laminas is *N*



Laminate strains

- Following the assumption that laminate strains are linearly related to the distance from the midplane as
  - $\epsilon_{xx} = \epsilon_{xx}{}^\circ + zk_{xx}$
  - $\epsilon_{yy} = \epsilon_{yy}{}^\circ + zk_{yy}$
  - $\gamma_{xy} = \gamma_{xy}{}^\circ + zk_{xy}$
- Definitions of the variables
  - $\epsilon_{xx}{}^\circ, \epsilon_{yy}{}^\circ$ = midplane normal strains in the laminate
  - $\gamma_{xy}{}^\circ$ = midplane shear strain in the laminate
  - k_{xx}, k_{yy} = bending curvatures of the laminate
  - k_{xy} = twisting curvature of the laminate
  - z = distance from the midplane in the thickness direction

Laminate forces and moments



- Applied force and moment resultant on a laminate are related to the midplane strains and curvatures by the following equations:

$$N_{xx} = A_{11}\epsilon_{xx}{}^\circ + A_{12}\epsilon_{yy}{}^\circ + A_{16}\gamma_{xy}{}^\circ + B_{11}k_{xx} + B_{12}k_{yy} + B_{16}k_{xy}$$
$$N_{yy} = A_{12}\epsilon_{xx}{}^\circ + A_{22}\epsilon_{yy}{}^\circ + A_{26}\gamma_{xy}{}^\circ + B_{12}k_{xx} + B_{22}k_{yy} + B_{26}k_{xy}$$
$$N_{xy} = A_{16}\epsilon_{xx}{}^\circ + A_{26}\epsilon_{yy}{}^\circ + A_{66}\gamma_{xy}{}^\circ + B_{16}k_{xx} + B_{26}k_{yy} + B_{66}k_{xy}$$
$$M_{xx} = B_{11}\epsilon_{xx}{}^\circ + B_{12}\epsilon_{yy}{}^\circ + B_{16}\gamma_{xy}{}^\circ + D_{11}k_{xx} + D_{12}k_{yy} + D_{16}k_{xy}$$
$$M_{yy} = B_{12}\epsilon_{xx}{}^\circ + B_{22}\epsilon_{yy}{}^\circ + B_{26}\gamma_{xy}{}^\circ + D_{12}k_{xx} + D_{22}k_{yy} + D_{26}k_{xy}$$
$$M_{xy} = B_{16}\epsilon_{xx}{}^\circ + B_{26}\epsilon_{yy}{}^\circ + B_{66}\gamma_{xy}{}^\circ + D_{16}k_{xx} + D_{26}k_{yy} + D_{66}k_{xy}$$

Laminate forces and moments – ABBD matrix

$$\begin{bmatrix} N_x \\ N_{yy} \\ N_{xy} \\ \hline M_{xx} \\ M_{yy} \\ M_{xy} \end{bmatrix} = \left[\begin{array}{ccc|ccc} A_{11} & A_{12} & A_{16} & B_{11} & B_{12} & B_{16} \\ A_{12} & A_{22} & A_{26} & B_{12} & B_{22} & B_{26} \\ A_{16} & A_{26} & A_{66} & B_{16} & B_{26} & B_{66} \\ \hline B_{11} & B_{12} & B_{16} & D_{11} & D_{12} & D_{16} \\ B_{12} & B_{22} & B_{26} & D_{12} & D_{22} & D_{26} \\ B_{16} & B_{26} & B_{66} & D_{16} & D_{26} & D_{66} \end{array}\right] \begin{bmatrix} \varepsilon_x^0 \\ \varepsilon_y^0 \\ \gamma_{xy}^0 \\ \hline K_{xx} \\ K_{yy} \\ K_{xy} \end{bmatrix}$$

Matrix notation

$$\begin{bmatrix} N_{\mathrm{xx}} \\ N_{\mathrm{yy}} \\ N_{\mathrm{xy}} \end{bmatrix} = [A] \begin{bmatrix} \epsilon_{xx}{}^\circ \\ \epsilon_{yy}{}^\circ \\ \gamma_{xy}{}^\circ \end{bmatrix} + [B] \begin{bmatrix} k_{xx} \\ k_{yy} \\ k_{xy} \end{bmatrix} \qquad \begin{bmatrix} M_{\mathrm{xx}} \\ M_{\mathrm{yy}} \\ M_{\mathrm{xy}} \end{bmatrix} = [B] \begin{bmatrix} \epsilon_{xx}{}^\circ \\ \epsilon_{yy}{}^\circ \\ \gamma_{xy}{}^\circ \end{bmatrix} + [D] \begin{bmatrix} k_{xx} \\ k_{yy} \\ k_{xy} \end{bmatrix}$$

- N_{xx} = normal force resultant in the *x* direction (per unit width)
- N_{yy} = normal force resultant in the *y* direction (per unit width)
- N_{xy} = shear force resultant (per unit width)
- M_{xx} = bending moment resultant in the *yz* plane (per unit width)
- M_{yy} = bending moment resultant in the *xz* plane (per unit width)
- M_{xy} = twisting moment (torsion) resultant (per unit width)

Stiffness matrix

$$[A] = \begin{bmatrix} A_{11} & A_{12} & A_{16} \\ A_{12} & A_{22} & A_{26} \\ A_{16} & A_{26} & A_{66} \end{bmatrix}$$

- Extensional stiffness matrix for the laminate (unit: N/m or lb/in.)

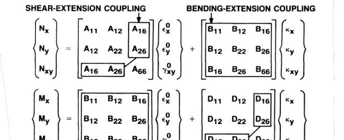
$$[B] = \begin{bmatrix} B_{11} & B_{12} & B_{16} \\ B_{12} & B_{22} & B_{26} \\ B_{16} & B_{26} & B_{66} \end{bmatrix}$$

- Coupling stiffness matrix for the laminate (unit: N or lb)

$$[D] = \begin{bmatrix} D_{11} & D_{12} & D_{16} \\ D_{12} & D_{22} & D_{26} \\ D_{16} & D_{26} & D_{66} \end{bmatrix}$$

- Bending stiffness matrix for the laminate (unit: Nm or lb in.)

Some physical insights


SHEAR-EXTENSION COUPLING
BENDING-EXTENSION COUPLING
N_x
N_y
N_{xy}
A_{11} A_{12} A_{16}
A_{12} A_{22} A_{26}
A_{16} A_{26} A_{66}
ϵ_x^0
ϵ_y^0
γ_{xy}^0
B_{11} B_{12} B_{16}
B_{12} B_{22} B_{26}
B_{16} B_{26} B_{66}
κ_x
κ_y
κ_{xy}
M_x
M_y
M_{xy}
D_{11} D_{12} D_{16}
D_{12} D_{22} D_{26}
D_{16} D_{26} D_{66}
BENDING-EXTENSION COUPLING
BEND-TWIST COUPLING

Elements in stiffness matrices

- Calculating moment and forces through integration of stresses in each layer (Jones Chapter 4 for more details), we obtain the following
- The elements in $[A], [B], [D]$ matrices are calculated from:
- $A_{mn} = \sum_{j=1}^{N} \overline{(Q_{mn})_j}(h_j - h_{j-1})$
- $B_{mn} = \frac{1}{2}\sum_{j=1}^{N} \overline{(Q_{mn})_j}(h_j^2 - h_{j-1}^2)$
- $D_{mn} = \frac{1}{3}\sum_{j=1}^{N} \overline{(Q_{mn})_j}(h_j^3 - h_{j-1}^3)$
- N = total number of laminas in the laminate
- $\overline{(Q_{mn})_j}$ = elements in the $[\bar{Q}]$ matrix of the *j*th lamina
- h_{j-1} = distance from the midplane to the top of the *j*th lamina
- h_j = distance from the midplane to the bottom of the *j*th lamina
- h_j is positive below the midplane and negative above the midplane

Things to note

- The elements of the stiffness matrices $[A]$, $[B]$, and $[D]$ are functions of the elastic properties of each lamina and its location with respect to the midplane of the laminate
- For a symmetric laminate, $[B] = [0]$ and there is no extension-bending coupling
- If $[B]$ is a nonzero matrix, a normal force will create extension as well as bending twisting curvatures.
- Balanced laminate is typically studied in which for every lamina of $+\theta$ orientation, there is an identical (equal in thickness and material) lamina of $-\theta$ orientation. For example, [0/30/-30/-30/30/0] is a balanced symmetric laminate. Balanced laminates do not exhibit shear-extension coupling.

Midplane strains and curvatures

- If the normal force and moment resultants acting on a laminate are known, its midplane strains and curvatures can be calculated

$$\begin{bmatrix} \epsilon_{xx}{}^\circ \\ \epsilon_{yy}{}^\circ \\ \gamma_{xy}{}^\circ \end{bmatrix} = [A_1] \begin{bmatrix} N_{\mathrm{xx}} \\ N_{\mathrm{yy}} \\ N_{\mathrm{xy}} \end{bmatrix} + [B_1] \begin{bmatrix} M_{\mathrm{xx}} \\ M_{\mathrm{yy}} \\ M_{\mathrm{xy}} \end{bmatrix} \qquad \begin{bmatrix} k_{xx} \\ k_{yy} \\ k_{xy} \end{bmatrix} = [C_1] \begin{bmatrix} N_{\mathrm{xx}} \\ N_{\mathrm{yy}} \\ N_{\mathrm{xy}} \end{bmatrix} + [D_1] \begin{bmatrix} M_{\mathrm{xx}} \\ M_{\mathrm{yy}} \\ M_{\mathrm{xy}} \end{bmatrix}$$

- For a symmetric laminate, equations for midplane strains and curvatures become

$$\begin{bmatrix} \epsilon_{xx}{}^\circ \\ \epsilon_{yy}{}^\circ \\ \gamma_{xy}{}^\circ \end{bmatrix} = [A^{-1}] \begin{bmatrix} N_{\mathrm{xx}} \\ N_{\mathrm{yy}} \\ N_{\mathrm{xy}} \end{bmatrix} \qquad \begin{bmatrix} k_{xx} \\ k_{yy} \\ k_{xy} \end{bmatrix} = [D^{-1}] \begin{bmatrix} M_{\mathrm{xx}} \\ M_{\mathrm{yy}} \\ M_{\mathrm{xy}} \end{bmatrix}$$

Elastic properties of a balanced symmetric laminate

Elastic properties of a balanced symmetric laminate: For a balanced symmetric laminate, the extensional stiffness matrix is

$$[A] = \begin{bmatrix} A_{11} & A_{12} & 0 \\ A_{12} & A_{22} & 0 \\ 0 & 0 & A_{66} \end{bmatrix}$$

and the coupling stiffness matrix $[B] = [0]$.
The inverse of the $[A]$ matrix is

$$[A^{-1}] = \frac{1}{A_{11}A_{22} - A_{12}^2} \begin{bmatrix} A_{22} & -A_{12} & 0 \\ -A_{12} & A_{11} & 0 \\ 0 & 0 & \dfrac{(A_{11}A_{22} - A_{12}^2)}{A_{66}} \end{bmatrix}.$$

Elastic properties

$$\begin{bmatrix} \varepsilon_{xx}^\circ \\ \varepsilon_{yy}^\circ \\ \gamma_{xy}^\circ \end{bmatrix} = \frac{1}{A_{11}A_{22} - A_{12}^2} \begin{bmatrix} A_{22} & -A_{12} & 0 \\ -A_{12} & A_{11} & 0 \\ 0 & 0 & \dfrac{(A_{11}A_{22} - A_{12}^2)}{A_{66}} \end{bmatrix} \begin{bmatrix} N_{xx} \\ N_{yy} \\ N_{xy} \end{bmatrix}$$

- Let us assume that the laminate is subjected to a uniaxial tensile stress σ_{xx} in the *x* direction, and both σ_{yy} and τ_{xy} are zero
- If the laminate thickness is h, the tensile force per unit width in the *x* direction $N_{\mathrm{xx}} = h\sigma_{xx}$, $N_{yy} = 0$, and $N_{xy} = 0$

$$\varepsilon_{xx}^\circ = \frac{A_{22}}{A_{11}A_{22} - A_{12}^2} h\sigma_{xx}, \quad \varepsilon_{yy}^\circ = -\frac{A_{12}}{A_{11}A_{22} - A_{12}^2} h\sigma_{xx}, \quad \gamma_{xy}^\circ = 0,$$

$$\boxed{E_{xx} = \frac{\sigma_{xx}}{\varepsilon_{xx}^\circ} = \frac{A_{11}A_{22} - A_{12}^2}{hA_{22}}, \quad \nu_{xy} = -\frac{\varepsilon_{yy}^\circ}{\varepsilon_{xx}^\circ} = \frac{A_{12}}{A_{22}}.}$$

Elastic properties

In turn, applying N_{yy} and N_{xy} separately, we can determine

$$E_{yy} = \frac{A_{11}A_{22} - A_{12}^2}{hA_{11}},$$

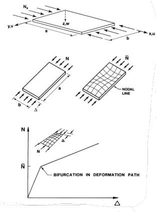
$$\nu_{yx} = \frac{A_{12}}{A_{11}} \left(\text{which is the same as } \nu_{xy}\frac{E_{yy}}{E_{xx}}\right),$$

and

$$G_{xy} = \frac{A_{66}}{h}.$$

Buckling of simply supported laminated plates

- Simply supported: $w = 0$ at $x = 0, x = a$ and $y = 0, y = b$
- Two-dimensional wavy nature (See the middle figure)
- Bifurcation: when the critical buckling load $\bar{N}$ is reached, the plate can sustain higher loads with decreased stiffness.



Buckling of specially orthotropic simply supported laminated plates under uniform in-plane load

- Specially orthotropic:

The laminates consists solely of $A_{11}, A_{12}, A_{22}, A_{66}, D_{11}, D_{12}, D_{22}, D_{66}$

- Buckling differential equation:

$$D_{11}\frac{\partial^4 w}{\partial x^4} + 2(D_{12} + 2D_{66})\frac{\partial^4 w}{\partial x^2 \partial y^2} + D_{22}\frac{\partial^4 w}{\partial y^4} + \bar{N}\frac{\partial^2 w}{\partial x^2} = 0$$
$$x = 0, a: \; w = 0, \qquad M_x = -D_{11}\frac{\partial^2 w}{\partial x^2} - D_{12}\frac{\partial^2 w}{\partial y^2} = 0$$
$$y = 0, b: \; w = 0, \qquad M_y = -D_{12}\frac{\partial^2 w}{\partial x^2} - D_{22}\frac{\partial^2 w}{\partial y^2} = 0$$

where w is the displacement of the mid-plane in z direction, M_x and M_y are the moment per unit width in x and y directions.

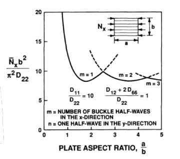
- General solution:

$$w = W_{mn}\sin\frac{m\pi x}{a}\sin\frac{n\pi y}{b}$$

where W_{mn} is the magnitude coefficient and m and n represents the number of buckle half wavelengths in x and y directions.

Buckling of specially orthotropic simply supported laminated plates under uniform in-plane load

- The geometry effect (aspect ratio a/b) on the critical buckling load (homework):



- Read section 5.4 in Jones book for the buckling analysis of more types of laminates

Bending of a symmetric laminate beam

Bending of a balanced symmetric laminate beam specimen: For a balanced symmetric laminate, $[B] = [0]$.

$$[D] = \begin{bmatrix} D_{11} & D_{12} & D_{16} \\ D_{12} & D_{22} & D_{26} \\ D_{16} & D_{26} & D_{66} \end{bmatrix},$$

$$[D^{-1}] = \frac{1}{D_0}\begin{bmatrix} D_{11}^\circ & D_{12}^\circ & D_{16}^\circ \\ D_{12}^\circ & D_{22}^\circ & D_{26}^\circ \\ D_{16}^\circ & D_{26}^\circ & D_{66}^\circ \end{bmatrix},$$

where

$D_0 = D_{11}(D_{22}D_{66} - D_{26}^2) - D_{12}(D_{12}D_{66} - D_{16}D_{26}) + D_{16}(D_{12}D_{26} - D_{22}D_{16})$

$D_{11}^\circ = (D_{22}D_{66} - D_{26}^2)$

$D_{12}^\circ = -(D_{12}D_{66} - D_{16}D_{26})$

$D_{16}^\circ = (D_{12}D_{26} - D_{22}D_{16})$

$D_{22}^\circ = (D_{11}D_{66} - D_{16}^2)$

$D_{26}^\circ = -(D_{11}D_{26} - D_{12}D_{16})$

$D_{66}^\circ = (D_{11}D_{12} - D_{12}^2)$

Bending of a symmetric laminate beam

- If a bending moment is applied in the *yz* plane so that M_{xx} is present and $M_{\mathrm{yy}} = M_{\mathrm{xy}} = 0$, the specimen curvatures can be obtained

$$k_{xx} = \frac{D_{11}^\circ}{D_0}M_{xx}, \quad k_{yy} = \frac{D_{12}^\circ}{D_0}M_{xx}, \quad k_{xy} = \frac{D_{16}^\circ}{D_0}M_{xx}.$$

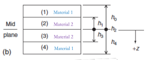
$$\boxed{E_{\mathrm{fl}} = \frac{M_{xx}}{k_{xx}I}}$$

E_{fl} is flexural (bending) modulus

Layered homogeneous, isotropic material case

- Variables in this case

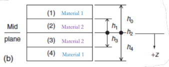
h_j - Distance from the midplane to the bottom of the j^{th} lamina
E_j - Young's Modulus of the j^{th} lamina
ν_j - Poisson's Ratio of the j^{th} lamina
G_j - Shear Modulus of the j^{th} lamina



Assume layer 1 and layer 4 are same material, layer 2 and layer 3 are same material

$$Q_{11} = \frac{E_{11}}{1 - \nu_{12}\nu_{21}} = \frac{E}{1 - \nu^2}$$

$$Q_{22} = \frac{E_{22}}{1 - \nu_{12}\nu_{21}} = \frac{E}{1 - \nu^2}$$



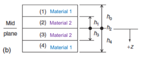
$$Q_{12} = Q_{21} = \frac{\nu_{12}E_{22}}{1 - \nu_{12}\nu_{21}} = \frac{\nu E}{1 - \nu^2}$$

$$Q_{66} = G_{12} = G$$

$$[Q]_j = \begin{bmatrix} \frac{E_j}{1-\nu_j^2} & \frac{\nu_j E_j}{1-\nu_j^2} & 0 \\ \frac{\nu_j E_j}{1-\nu_j^2} & \frac{E_j}{1-\nu_j^2} & 0 \\ 0 & 0 & G_j \end{bmatrix}$$

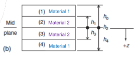
For a homogeneous and isotropic material:

$$[\mathrm{Q}]_j = [\mathrm{Q}]_j$$



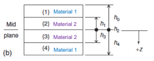
$$h_3 = -h_1, h_4 = -h_0, h_2 = 0$$
$$h_1 - h_0 = h_4 - h_3, h_2 - h_1 = h_3 - h_2$$
$$[\bar{Q}]_1 = [\bar{Q}]_4, [\bar{Q}]_2 = [\bar{Q}]_3$$

$$A_{mn} = \sum_{j=1}^{N}\overline{(Q_{mn})_j}(h_j - h_{j-1})$$
$$= \{\overline{(Q_{mn})_1}(h_1 - h_0) + \overline{(Q_{mn})_4}(h_4 - h_3)\} + \{\overline{(Q_{mn})_2}(h_2 - h_1) + \overline{(Q_{mn})_3}(h_3 - h_2)\}$$
$$= 2\overline{(Q_{mn})_1}(h_1 - h_0) + 2\overline{(Q_{mn})_2}(h_2 - h_1)$$



$$h_3 = -h_1, h_4 = -h_0, h_2 = 0$$
$$-h_1^2 + h_0^2 = h_4^2 - h_3^2, -h_2^2 + h_1^2 = h_3^2 - h_2^2$$
$$[\bar{Q}]_1 = [\bar{Q}]_4, [\bar{Q}]_2 = [\bar{Q}]_3$$

$$B_{mn} = \frac{1}{2}\sum_{j=1}^{N}\overline{(Q_{mn})_j}(h_j^2 - h_{j-1}^2)$$
$$= \frac{1}{2}\{\overline{(Q_{mn})_1}(h_1^2 - h_0^2) + \overline{(Q_{mn})_4}(h_4^2 - h_3^2)\} + \frac{1}{2}\{\overline{(Q_{mn})_2}(h_2^2 - h_1^2) + \overline{(Q_{mn})_3}(h_3^2 - h_2^2)\}$$
$$= 0 + 0 = 0$$



$$h_3 = -h_1, h_4 = -h_0, h_2 = 0$$
$$h_1^3 - h_0^3 = h_4^3 - h_3^3, h_2^3 - h_1^3 = h_3^3 - h_2^3$$
$$[\bar{Q}]_1 = [\bar{Q}]_4, [\bar{Q}]_2 = [\bar{Q}]_3$$

$$D_{mn} = \frac{1}{3}\sum_{j=1}^{N}\overline{(Q_{mn})_j}(h_j^3 - h_{j-1}^3)$$
$$= \frac{1}{3}\{\overline{(Q_{mn})_1}(h_1^3 - h_0^3) + \overline{(Q_{mn})_4}(h_4^3 - h_3^3)\} + \frac{1}{3}\{\overline{(Q_{mn})_2}(h_2^3 - h_1^3) + \overline{(Q_{mn})_3}(h_3^3 - h_2^3)\}$$
$$= \frac{2}{3}\overline{(Q_{mn})_1}(h_1^3 - h_0^3) + \frac{2}{3}\overline{(Q_{mn})_2}(h_2^3 - h_1^3)$$

Solution – calculate $[A]$, $[B]$, and $[D]$ matrices

- $[A] = 2[\bar{Q}]_1(h_1 - h_0) + 2[\bar{Q}]_2(h_2 - h_1)$

$$= 2(h_1 - h_0)\begin{bmatrix} \frac{E_1}{1-\nu_1^2} & \frac{\nu_1 E_1}{1-\nu_1^2} & 0 \\ \frac{\nu_1 E_1}{1-\nu_1^2} & \frac{E_1}{1-\nu_1^2} & 0 \\ 0 & 0 & G_1 \end{bmatrix} + 2(h_2 - h_1)\begin{bmatrix} \frac{E_2}{1-\nu_2^2} & \frac{\nu_2 E_2}{1-\nu_2^2} & 0 \\ \frac{\nu_2 E_2}{1-\nu_2^2} & \frac{E_2}{1-\nu_2^2} & 0 \\ 0 & 0 & G_2 \end{bmatrix}$$

- $[B] = [0]$
- $[D] = \frac{2}{3}[\bar{Q}]_1(h_1^3 - h_0^3) + \frac{2}{3}[\bar{Q}]_2(h_2^3 - h_1^3)$

$$= \frac{2}{3}(h_1^3 - h_0^3)\begin{bmatrix} \frac{E_1}{1-\nu_1^2} & \frac{\nu_1 E_1}{1-\nu_1^2} & 0 \\ \frac{\nu_1 E_1}{1-\nu_1^2} & \frac{E_1}{1-\nu_1^2} & 0 \\ 0 & 0 & G_1 \end{bmatrix} + \frac{2}{3}(h_2^3 - h_1^3)\begin{bmatrix} \frac{E_2}{1-\nu_2^2} & \frac{\nu_2 E_2}{1-\nu_2^2} & 0 \\ \frac{\nu_2 E_2}{1-\nu_2^2} & \frac{E_2}{1-\nu_2^2} & 0 \\ 0 & 0 & G_2 \end{bmatrix}$$

Lamina strains and stresses due to applied loads

- Knowing the midplane strains and curvatures for the laminate, strains at the midplane at each lamina can be calculated using

$$\begin{bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \gamma_{xy} \end{bmatrix}_j = \begin{bmatrix} \varepsilon_{xx}^\circ \\ \varepsilon_{yy}^\circ \\ \gamma_{xy}^\circ \end{bmatrix} + z_j \begin{bmatrix} k_{xx} \\ k_{yy} \\ k_{xy} \end{bmatrix},$$

- where z_j is the distance from the laminate midplane to the midplane of the *j*th lamina
- Stresses in the *j*th lamina can be calculated using its stiffness matrix

$$\begin{bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \tau_{xy} \end{bmatrix}_j = [\bar{Q}_{\mathrm{mn}}]_j \begin{bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \gamma_{xy} \end{bmatrix}_j = [\bar{Q}_{\mathrm{mn}}]_j \begin{bmatrix} \varepsilon_{xx}^\circ \\ \varepsilon_{yy}^\circ \\ \gamma_{xy}^\circ \end{bmatrix} + z_j[\bar{Q}_{\mathrm{mn}}]_j \begin{bmatrix} k_{xx} \\ k_{yy} \\ k_{xy} \end{bmatrix}.$$

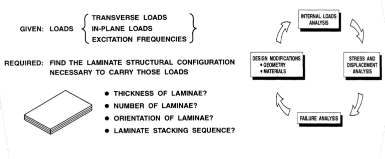
Application of failure criterions

- Using stress transformation matrix, we may compute the longitudinal, transverse, and shear stresses in the 1-2 directions

	45° Layer (MPa)	**−45° Layer (MPa)**
σ_{11}	6.255	6.255
σ_{22}	2.075	2.075
τ_{12}	−4.165	4.165

Apply failure criterion to figure out if any plies have failed

Beauty of composites



Complexity of composites brings about exciting opportunities to optimize

Lamina strains and stresses due to thermal effects

- Since laminates are typically fabricated at high temperature, they can develop significant thermally induced stresses due to the manufacturing process
- For a 0° lamina (in plane stress), in the presence of a temperature change ΔT, the thermal strains has the form

$$\begin{bmatrix} \epsilon_1^t \\ \epsilon_2^t \\ \gamma_{12}^t \end{bmatrix} = \Delta T \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ 0 \end{bmatrix}$$

- where α_1 and α_2 are the coefficients of thermal expansion, and there is no thermal shear strain
- This form is usually valid if ΔT is not too large (compared to the current temperature) so that ϵ^t can be approximated as linear functions of ΔT
- Jones book Section 4.5.3. or Mallick's book section 3.3.2.7

Lamina strains and stresses due to thermal effects

- When stress is present, we assume the strains to be the sum of these due to stress and thermal effects

$$\begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \gamma_{12} \end{bmatrix} = [S]\begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \tau_{12} \end{bmatrix} + \Delta T\begin{bmatrix} \alpha_1 \\ \alpha_2 \\ 0 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \tau_{12} \end{bmatrix} = [Q]\begin{bmatrix} \epsilon_1 - \alpha_1\Delta T \\ \epsilon_2 - \alpha_2\Delta T \\ \gamma_{12} \end{bmatrix}$$

- For a θ-oriented lamina, the stress-strain relation in global x-y coordinates becomes

$$\begin{bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \tau_{xy} \end{bmatrix} = [\bar{Q}(\theta)]\begin{bmatrix} \epsilon_{xx} - \alpha_{xx}\Delta T \\ \epsilon_{yy} - \alpha_{yy}\Delta T \\ \gamma_{xy} - \alpha_{xy}\Delta T \end{bmatrix}$$

where

$$\alpha_{xx} = \alpha_1\cos^2\theta + \alpha_2\sin^2\theta$$
$$\alpha_{yy} = \alpha_2\cos^2\theta + \alpha_1\sin^2\theta$$
$$\alpha_{xy} = 2\sin\theta\cos\theta(\alpha_1 - \alpha_2)$$

Laminate strains and stresses due to thermal effects

- For a laminate, including thermal effects, the stress in the *j*th lamina is

$$\begin{bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \tau_{xy} \end{bmatrix}_j = [\bar{Q}_{mn}]_j\left(\begin{bmatrix} \epsilon_{xx} \\ \epsilon_{yy} \\ \gamma_{xy} \end{bmatrix}_j - \Delta T\begin{bmatrix} \alpha_{xx} \\ \alpha_{yy} \\ \alpha_{xy} \end{bmatrix}_j\right)$$

- Therefore

$$\begin{bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \tau_{xy} \end{bmatrix}_j = [\bar{Q}_{mn}]_j\begin{bmatrix} \epsilon_{xx}^\circ \\ \epsilon_{yy}^\circ \\ \gamma_{xy}^\circ \end{bmatrix} + z_j[\bar{Q}_{mn}]_j\begin{bmatrix} k_{xx} \\ k_{yy} \\ k_{xy} \end{bmatrix} - \Delta T[\bar{Q}_{mn}]_j\begin{bmatrix} \alpha_{xx} \\ \alpha_{yy} \\ \alpha_{xy} \end{bmatrix}_j$$

Laminate strains and stresses due to thermal effects

- Resultant forces and moments

$$\begin{bmatrix} N_{xx} \\ N_{yy} \\ N_{xy} \end{bmatrix} = [A]\begin{bmatrix} \epsilon_{xx}^\circ \\ \epsilon_{yy}^\circ \\ \gamma_{xy}^\circ \end{bmatrix} + [B]\begin{bmatrix} k_{xx} \\ k_{yy} \\ k_{xy} \end{bmatrix} - [N^t]$$

$$\begin{bmatrix} M_{xx} \\ M_{yy} \\ M_{xy} \end{bmatrix} = [B]\begin{bmatrix} \epsilon_{xx}^\circ \\ \epsilon_{yy}^\circ \\ \gamma_{xy}^\circ \end{bmatrix} + [D]\begin{bmatrix} k_{xx} \\ k_{yy} \\ k_{xy} \end{bmatrix} - [M^t]$$

where

$$[N^t] = \sum_{j=1}^{N}\overline{(Q_{mn})_j}[\alpha]_j(z_j - z_{j-1})\Delta T$$

$$[M^t] = \frac{1}{2}\sum_{j=1}^{N}\overline{(Q_{mn})_j}[\alpha]_j(z_j^2 - z_{j-1}^2)\Delta T$$

Reduced stiffness for plane stress state

Inverting the previous equation:

$$\begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \tau_{12} \end{bmatrix} = \begin{bmatrix} Q_{11} & Q_{12} & 0 \\ Q_{12} & Q_{22} & 0 \\ 0 & 0 & Q_{66} \end{bmatrix}\begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \gamma_{12} \end{bmatrix}$$

$$Q_{11} = \frac{S_{22}}{S_{11}S_{22} - S_{12}^2} = \frac{E_1}{1 - \nu_{12}\nu_{21}} \qquad Q_{22} = \frac{S_{11}}{S_{11}S_{22} - S_{12}^2} = \frac{E_2}{1 - \nu_{12}\nu_{21}}$$
$$Q_{12} = \frac{S_{12}}{S_{11}S_{22} - S_{12}^2} = \frac{\nu_{12}E_2}{1 - \nu_{12}\nu_{21}} = \frac{\nu_{21}E_1}{1 - \nu_{12}\nu_{21}} \qquad Q_{66} = G_{12}$$

- $\mathrm{Q_{ij}}$ are the so-called reduced stiffness for a plane stress state in x-y plane

Properties as function of fiber orientation

$$\begin{bmatrix} \epsilon_x \\ \epsilon_y \\ \gamma_{xy} \end{bmatrix} = [R][\mathrm{T}]^{-1}[R]^{-1}\begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \gamma_{12} \end{bmatrix} = [T]^T\begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \gamma_{12} \end{bmatrix} = [T]^T[S][T]\begin{bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{bmatrix} = [\bar{\mathrm{S}}]\begin{bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{bmatrix}$$

$$[\bar{\mathrm{S}}] = [T]^T[S][T] = [\bar{Q}]^{-1}$$

$$\boxed{\begin{aligned} \overline{S_{11}} &= \frac{1}{E_{xx}} = \frac{\cos^4\theta}{E_{11}} + \frac{\sin^4\theta}{E_{22}} + \frac{1}{4}\left(\frac{1}{G_{12}} - \frac{2\nu_{12}}{E_{11}}\right)\sin^2 2\theta \\ \overline{S_{22}} &= \frac{1}{E_{yy}} = \frac{\sin^4\theta}{E_{11}} + \frac{\cos^4\theta}{E_{22}} + \frac{1}{4}\left(\frac{1}{G_{12}} - \frac{2\nu_{12}}{E_{11}}\right)\sin^2 2\theta \\ \overline{S_{66}} &= \frac{1}{G_{xy}} = \frac{1}{E_{11}} + \frac{2\nu_{12}}{E_{11}} + \frac{1}{E_{22}} - \left(\frac{1}{E_{11}} + \frac{2\nu_{12}}{E_{11}} + \frac{1}{E_{22}} - \frac{1}{G_{12}}\right)\cos^2 2\theta \\ \overline{S_{12}} &= -\nu_{xy}/E_{xx} = -[\frac{\nu_{12}}{E_{11}} - \frac{1}{4}\left(\frac{1}{E_{11}} + \frac{2\nu_{12}}{E_{11}} + \frac{1}{E_{22}} - \frac{1}{G_{12}}\right)\sin^2 2\theta] \\ \nu_{yx} &= \frac{E_{yy}}{E_{xx}}\nu_{xy} \end{aligned}}$$

Stress transformation from 1-2 to x-y directions

Matrix form

$$\begin{bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{bmatrix} = \begin{bmatrix} \cos^2\theta & \sin^2\theta & -2\sin\theta\cos\theta \\ \sin^2\theta & \cos^2\theta & 2\sin\theta\cos\theta \\ \sin\theta\cos\theta & -\sin\theta\cos\theta & \cos^2\theta - \sin^2\theta \end{bmatrix}\begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \tau_{12} \end{bmatrix} = [\mathrm{T}]^{-1}\begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \tau_{12} \end{bmatrix}$$

$$\begin{bmatrix} \epsilon_x \\ \epsilon_y \\ \frac{\gamma_{xy}}{2} \end{bmatrix} = \begin{bmatrix} \cos^2\theta & \sin^2\theta & -2\sin\theta\cos\theta \\ \sin^2\theta & \cos^2\theta & 2\sin\theta\cos\theta \\ \sin\theta\cos\theta & -\sin\theta\cos\theta & \cos^2\theta - \sin^2\theta \end{bmatrix}\begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \frac{\gamma_{12}}{2} \end{bmatrix} = [\mathrm{T}]^{-1}\begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \frac{\gamma_{12}}{2} \end{bmatrix}$$

$$[\mathrm{T}]^{-1} = \begin{bmatrix} \cos^2\theta & \sin^2\theta & -2\sin\theta\cos\theta \\ \sin^2\theta & \cos^2\theta & 2\sin\theta\cos\theta \\ \sin\theta\cos\theta & -\sin\theta\cos\theta & \cos^2\theta - \sin^2\theta \end{bmatrix}$$

Stress-strain relations in x-y coordinates

- Question: Given the stress-strain relations in the principal material coordinates, what are the stress-strain relations in x-y coordinates?

We may introduce a simple matrix

$$[\mathrm{R}] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \qquad \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \gamma_{12} \end{bmatrix} = [\mathrm{R}]\begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \frac{\gamma_{12}}{2} \end{bmatrix} \text{ and } \begin{bmatrix} \epsilon_x \\ \epsilon_y \\ \gamma_{xy} \end{bmatrix} = [\mathrm{R}]\begin{bmatrix} \epsilon_x \\ \epsilon_y \\ \frac{\gamma_{xy}}{2} \end{bmatrix}$$

Removing awkward factors of ½ and 2

$$\begin{bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{bmatrix} = [\mathrm{T}]^{-1}\begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \tau_{12} \end{bmatrix} = [\mathrm{T}]^{-1}[\mathrm{Q}]\begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \gamma_{12} \end{bmatrix} = [\mathrm{T}]^{-1}[\mathrm{Q}][\mathrm{R}][\mathrm{T}][\mathrm{R}]^{-1}\begin{bmatrix} \epsilon_x \\ \epsilon_y \\ \gamma_{xy} \end{bmatrix}$$

Full matrix

$$\begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \tau_{12} \end{bmatrix} = \begin{bmatrix} Q_{11} & Q_{12} & 0 \\ Q_{12} & Q_{22} & 0 \\ 0 & 0 & Q_{66} \end{bmatrix}\begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \gamma_{12} \end{bmatrix} = [\mathrm{Q}]\begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \gamma_{12} \end{bmatrix}$$

Use of reduced stiffness matrix

Transformed stiffness matrix

$$\begin{bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{bmatrix} = [\mathrm{T}]^{-1}[\mathrm{Q}][\mathrm{R}][\mathrm{T}][\mathrm{R}]^{-1}\begin{bmatrix} \epsilon_x \\ \epsilon_y \\ \gamma_{xy} \end{bmatrix}$$

Define $[\bar{\mathrm{Q}}] = [\mathrm{T}]^{-1}[\mathrm{Q}][\mathrm{R}][\mathrm{T}][\mathrm{R}]^{-1} = [\mathrm{T}]^{-1}[\mathrm{Q}][\mathrm{T}]^{-\mathrm{T}}$

$$\begin{bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{bmatrix} = [\bar{\mathrm{Q}}]\begin{bmatrix} \epsilon_x \\ \epsilon_y \\ \gamma_{xy} \end{bmatrix} = \begin{bmatrix} \overline{Q_{11}} & \overline{Q_{12}} & \overline{Q_{16}} \\ \overline{Q_{12}} & \overline{Q_{22}} & \overline{Q_{26}} \\ \overline{Q_{16}} & \overline{Q_{26}} & \overline{Q_{66}} \end{bmatrix}\begin{bmatrix} \epsilon_x \\ \epsilon_y \\ \gamma_{xy} \end{bmatrix}$$

- Where the bar over the $\overline{Q_{ij}}$ matrix denotes that we are dealing with the transformed reduced stiffnesses instead of the reduced stiffnesses, Q_{ij}

Expression of strains in terms of stresses

- Transformation of the stress-strain relations in principal material coordinates
- We want to express the material property constants (young's modulus, Poisson's ratio, shear modulus) in x-y directions in terms of the constants in fiber directions, i.e., 1-2 directions. So we may look for matrix $[\bar{S}]$ such that

$$\begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \gamma_{12} \end{bmatrix} = \begin{bmatrix} S_{11} & S_{12} & 0 \\ S_{12} & S_{22} & 0 \\ 0 & 0 & S_{66} \end{bmatrix}\begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \tau_{12} \end{bmatrix} \qquad \begin{bmatrix} \epsilon_x \\ \epsilon_y \\ \gamma_{xy} \end{bmatrix} = [\bar{\mathrm{S}}]\begin{bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{bmatrix} = \begin{bmatrix} \overline{S_{11}} & \overline{S_{12}} & \overline{S_{16}} \\ \overline{S_{12}} & \overline{S_{22}} & \overline{S_{26}} \\ \overline{S_{16}} & \overline{S_{26}} & \overline{S_{66}} \end{bmatrix}\begin{bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{bmatrix}$$

$$\overline{S_{11}} = \frac{1}{E_{xx}}, \; \overline{S_{12}} = -\frac{\nu_{xy}}{E_{xx}} = -\frac{\nu_{yx}}{E_{yy}}, \; \overline{S_{22}} = \frac{1}{E_{yy}}, \; \overline{S_{66}} = \frac{1}{G_{xy}}$$

Expression of strains in terms of stresses

$$\begin{bmatrix} \epsilon_x \\ \epsilon_y \\ \frac{\gamma_{xy}}{2} \end{bmatrix} = \begin{bmatrix} \cos^2\theta & \sin^2\theta & -2\sin\theta\cos\theta \\ \sin^2\theta & \cos^2\theta & 2\sin\theta\cos\theta \\ \sin\theta\cos\theta & -\sin\theta\cos\theta & \cos^2\theta - \sin^2\theta \end{bmatrix}\begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \frac{\gamma_{12}}{2} \end{bmatrix} = [\mathrm{T}]^{-1}\begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \frac{\gamma_{12}}{2} \end{bmatrix}$$

$$[\mathrm{T}] = \begin{bmatrix} \cos^2\theta & \sin^2\theta & 2\sin\theta\cos\theta \\ \sin^2\theta & \cos^2\theta & -2\sin\theta\cos\theta \\ -\sin\theta\cos\theta & \sin\theta\cos\theta & \cos^2\theta - \sin^2\theta \end{bmatrix}$$

$$\begin{bmatrix} \epsilon_x \\ \epsilon_y \\ \gamma_{xy} \end{bmatrix} = [R]\begin{bmatrix} \epsilon_x \\ \epsilon_y \\ \frac{\gamma_{xy}}{2} \end{bmatrix} = [R][\mathrm{T}]^{-1}\begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \frac{\gamma_{12}}{2} \end{bmatrix} = [R][\mathrm{T}]^{-1}[R]^{-1}\begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \gamma_{12} \end{bmatrix}$$

$$\begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \gamma_{12} \end{bmatrix} = \begin{bmatrix} S_{11} & S_{12} & 0 \\ S_{12} & S_{22} & 0 \\ 0 & 0 & S_{66} \end{bmatrix}\begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \tau_{12} \end{bmatrix} = [S]\begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \tau_{12} \end{bmatrix} = [S][T]\begin{bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{bmatrix}$$

Compliance matrix

Deformation response

Applied stress

$$\begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \gamma_{12} \end{bmatrix} = \begin{bmatrix} S_{11} & S_{12} & 0 \\ S_{12} & S_{22} & 0 \\ 0 & 0 & S_{66} \end{bmatrix}\begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \tau_{12} \end{bmatrix}$$

$$S_{11} = \frac{1}{E_{11}}, \; S_{12} = -\frac{\nu_{12}}{E_{11}} = -\frac{\nu_{21}}{E_{22}}, \; S_{22} = \frac{1}{E_{22}}, \; S_{66} = \frac{1}{G_{12}}$$